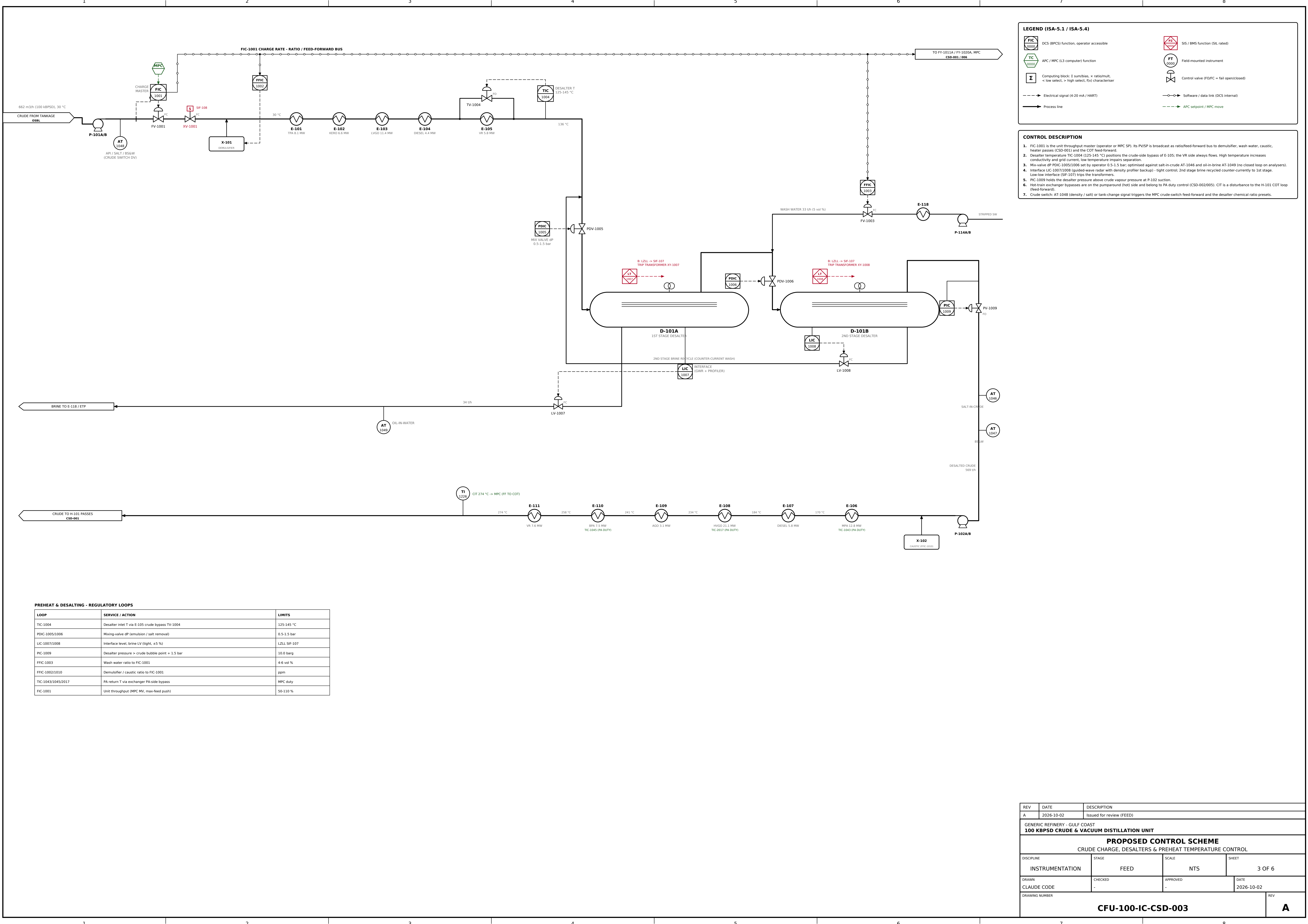




LEGEND (ISA-5.1 / ISA-5.4)

FIC
0000

DCS (BPCS) function, operator accessible

TC
0000

APC / MPC (L3 computer) function

Σ

Computing block: Σ sum/bias, × ratio/mult, < low select, > high select, (fx) characteriser

→

Electrical signal (4-20 mA / HART)

→

Process line

SIS
0000

SIS / BMS function (SIL rated)

FT
0000

Field-mounted instrument

CV

Control valve (FO/FC = fail open/closed)

—○—

Software / data link (DCS internal)

—○—

APC setpoint / MPC move

CONTROL DESCRIPTION

- FIC-1001 is the unit throughput master (operator or MPC SPI), its PV/SP is broadcast as ratio/feed-forward bus to demulsifier, wash water, caustic, heater passes (CSD-001) and the COT feed-forward.
- Desalter temperature TIC-1004 (125-145 °C) positions the crude-side bypass of E-105; the VR side always flows. High temperature increases conductivity and grid current, low temperature impairs separation.
- Mix-valve dP PDIC-1005/1006 set by operator 0.5-1.5 bar; optimised against salt-in-crude AT-1046 and oil-in-brine AT-1049 (no closed loop on analysers).
- Interface LIC-1007/1008 (guided-wave radar with density profiler backup) - tight control; 2nd stage brine recycled counter-currently to 1st stage.
- Low-low interface SIF-107 trips the transformers.
- PIC-1009 holds the desalter pressure above crude vapour pressure at P-102 suction.
- Hot-train exchanger bypasses are on the pumparound (hot) side and belong to PA duty control (CSD-002/005). CIT is a disturbance to the H-101 COT loop (feed-forward).
- Crude switch: AT-1048 (density / salt) or tank-change signal triggers the MPC crude-switch feed-forward and the desalter chemical ratio presets.

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST			
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PROPOSED CONTROL SCHEME			
CRUDE CHARGE, DESALTERS & PREHEAT TEMPERATURE CONTROL			
DISCIPLINE	STAGE	SCALE	SHEET
INSTRUMENTATION	FEED	NTS	3 OF 6
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-100-IC-CSD-003			A